EQ1

Sia $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$. Calcolare:

(a) $AX = B$ $\quad B = \begin{bmatrix} 3 & 1 & 2 \\ 6 & 2 & 4 \\ 2 & 1 & 1 \end{bmatrix}$

(b) $YA = C$ $\quad C = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix}$

Studiamo l'invertibilità di A

(1 Modo - MEG). Studiamo rango e cerchiamo contemporaneamente l'ipotetica inversa.

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 4 & 6 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array}\right] \begin{matrix} R_2 - 2R_1 \\ \longrightarrow \\ R_3 - R_1 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 0 & 0 & -2 & 1 & 0 \\ 0 & -1 & -2 & -1 & 0 & 1 \end{array}\right] \xrightarrow{R_3 \leftrightarrow R_2}$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -1 & -2 & -1 & 0 & 1 \\ 0 & 0 & 0 & -2 & 1 & 0 \end{array}\right]$$

$\rho(A) = 2 \Rightarrow A$ singolare $\Rightarrow \nexists A^{-1}$

(2 Modo - Determinanti)

Ci sono due righe proporzionali (1^a e 2^a) $\Rightarrow |A| = 0$.

Volendolo calcolare, es. 1^a riga:

$$|A| = 1 \cdot \begin{vmatrix} 4 & 6 \\ 1 & 1 \end{vmatrix} - 2 \begin{vmatrix} 2 & 6 \\ 1 & 1 \end{vmatrix} + 3 \begin{vmatrix} 2 & 4 \\ 1 & 1 \end{vmatrix} = (4-6) - 2(2-6) + 3(2-4) =$$

$$= -2 + 8 - 6 = 0 \Rightarrow A \text{ singolare} \Rightarrow \nexists A^{-1}.$$

Costruiamo le soluzioni:

(a) $AX = B$ con $B = \begin{bmatrix} 3 & 1 & 2 \\ 6 & 2 & 4 \\ 2 & 1 & 1 \end{bmatrix}$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 3 & 1 & 2 \\ 2 & 4 & 6 & 6 & 2 & 4 \\ 1 & 1 & 1 & 2 & 1 & 1 \end{array}\right] \begin{matrix} R_2 - 2R_1 \\ \longrightarrow \\ R_3 - R_1 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 3 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & -2 & -1 & 0 & -1 \end{array}\right] \xrightarrow{R_3 \leftrightarrow R_2}$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 3 & 1 & 2 \\ 0 & -1 & -2 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

x_1 x_2 x_3

$$\begin{cases} x_1+2x_2+3x_3=3 \\ -x_2-2x_3=-1 \\ x_3=\pi \end{cases} \quad \begin{cases} x_1'+2x_2'+3x_3'=1 \\ -x_2'-2x_3'=0 \\ x_3'=s \end{cases} \quad \begin{cases} x_1''+2x_2''+3x_3''=2 \\ -x_2''-2x_3''=-1 \\ x_3''=t \end{cases}$$

$$\begin{cases} x_1=-2+4\pi-3\pi+3 \\ x_2=1-2\pi \\ x_3=\pi \end{cases} \quad \begin{cases} x_1'=4s-3s+1 \\ x_2'=-2s \\ x_3'=s \end{cases} \quad \begin{cases} x_1''=4t-2-3t+2 \\ x_2''=1-2t \\ x_3''=t \end{cases}$$

$$\begin{cases} x_1=\pi+1 \\ x_2=1-2\pi \\ x_3=\pi \end{cases} \quad \begin{cases} x_1'=s+1 \\ x_2'=-2s \\ x_3'=s \end{cases} \quad \begin{cases} x_1''=t \\ x_2''=1-2t \\ x_3''=t \end{cases}$$

$$\Rightarrow A=\begin{bmatrix} x_1 & x_1' & x_1'' \\ x_2 & x_2' & x_2'' \\ x_3 & x_3' & x_3'' \end{bmatrix}=\begin{bmatrix} \pi+1 & s+1 & t \\ 1-2\pi & -2s & 1-2t \\ \pi & s & t \end{bmatrix} \quad \pi,s,t\in\mathbb{R}$$

test di prova:

$$AX=\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} \pi+1 & s+1 & t \\ 1-2\pi & -2s & 1-2t \\ \pi & s & t \end{bmatrix}=$$

$$=\begin{bmatrix} \pi+1+2-4\pi+3\pi & s+1-4s+3s & t+2-4t+3t \\ 2\pi+2+4-8\pi+6\pi & 2s+2-8s+6s & 2t+4-8t+6t \\ \pi+1+1-2\pi+\pi & s+s-2s+1+s & t+1-2t+t \end{bmatrix}$$

$$=\begin{bmatrix} 3 & 1 & 2 \\ 6 & 2 & 4 \\ 2 & 1 & 1 \end{bmatrix}=B$$

Da $YA = C$, passando alle Trasposte:

$$(YA)^T = C^T \Rightarrow A^T Y^T = C^T \Rightarrow \quad \text{con } C = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 1 \\ 3 & 6 & 1 \end{bmatrix} Y^T = \begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & 1 \\ 4 & 3 & 1 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 2 & 1 & 1 \\ 2 & 4 & 1 & 3 & 2 & 1 \\ 3 & 6 & 1 & 4 & 3 & 1 \end{array}\right] \begin{matrix} R_2 - 2R_1 \\ R_3 - 3R_1 \end{matrix} \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 2 & 1 & 1 \\ 0 & 0 & -1 & -1 & 0 & -1 \\ 0 & 0 & -2 & -2 & 0 & -2 \end{array}\right] R_3 - 2R_2 \longrightarrow$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 2 & 1 & 1 \\ 0 & 0 & -1 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow$$

$x_1 \quad x_2 \quad x_3$

$$\begin{cases} x_1 + 2x_2 + x_3 = 2 \\ -x_3 = -1 \\ x_2 = r \end{cases} \quad \begin{cases} x_1' + 2x_2' + x_3' = 1 \\ -x_3' = 0 \\ x_2' = s \end{cases} \quad \begin{cases} x_1'' + 2x_2'' + x_3'' = 1 \\ -x_3'' = -1 \\ x_2'' = t \end{cases}$$

$$\begin{cases} x_1 = 1 - 2r \\ x_2 = r \\ x_3 = 1 \end{cases} \quad \begin{cases} x_1' = 1 - 2s \\ x_2' = s \\ x_3' = 0 \end{cases} \quad \begin{cases} x_1'' = -2t \\ x_2'' = t \\ x_3'' = 1 \end{cases} \Rightarrow$$

$$Y^T = \begin{bmatrix} 1-2r & 1-2s & -2t \\ r & s & t \\ 1 & 0 & 1 \end{bmatrix} \Rightarrow Y = \begin{bmatrix} 1-2r & r & 1 \\ 1-2s & s & 0 \\ -2t & t & 1 \end{bmatrix}$$

test di prova:

$$YA = \begin{bmatrix} 1-2r & r & 1 \\ 1-2s & s & 0 \\ -2t & t & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix} =$$

$$= \begin{bmatrix} 1-2r+2r+1 & 2-4r+4r+1 & 3-6r+6r+1 \\ 1-2s+2s & 2-4s+4s & 3-6s+6s \\ -2t+2t+1 & -4t+4t+1 & -6t+6t+1 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix} = C$$

Sia $A=\begin{bmatrix}2&1\\1&1\end{bmatrix}$. Calcolare

(a) $AX=B \qquad B=\begin{bmatrix}3&5\\4&6\end{bmatrix}$

(b) $YA=C \qquad C=\begin{bmatrix}1&2\\3&4\end{bmatrix}$

Verifichiamo l'invertibilità di A

(1 Modo - MEG) Calcoliamo contemporaneamente rango ed eventuale inversa:

$$\left[\begin{array}{cc|cc}2&1&1&0\\1&1&0&1\end{array}\right] \xrightarrow{R_2-\frac{1}{2}R_1} \left[\begin{array}{cc|cc}2&1&1&0\\0&\frac{1}{2}&-\frac{1}{2}&1\end{array}\right] \xrightarrow[2R_2]{\frac{1}{2}R_1} \left[\begin{array}{cc|cc}1&\frac{1}{2}&\frac{1}{2}&0\\0&1&-1&2\end{array}\right]$$

$\exists A^{-1}$ perché $\pi(A)=2$

$$\xrightarrow{R_1-\frac{1}{2}R_2} \left[\begin{array}{cc|cc}1&0&1&-1\\0&1&-1&2\end{array}\right] \Rightarrow A^{-1}=\begin{bmatrix}1&-1\\-1&2\end{bmatrix}$$

(2 Modo - Matrice aggiunta)

$\det A=\begin{vmatrix}2&1\\1&1\end{vmatrix}=2-1=1\neq 0$ $\exists$ inversa (A non singolare)

$$A^{-1}=\frac{1}{|A|}(A^*)^T \text{ con } A^*=\begin{bmatrix}1&-1\\-1&2\end{bmatrix} \Rightarrow A^{-1}=\frac{1}{1}\begin{bmatrix}1&-1\\-1&2\end{bmatrix}^T=\begin{bmatrix}1&-1\\-1&2\end{bmatrix}$$

(a) $AX=B \Rightarrow (A^{-1}A)X=A^{-1}B$

$$X=\begin{bmatrix}1&-1\\-1&2\end{bmatrix}\begin{bmatrix}3&5\\4&6\end{bmatrix}=\begin{bmatrix}3-4&5-6\\-3+8&-5+12\end{bmatrix}=\begin{bmatrix}-1&-1\\5&7\end{bmatrix}$$

(b) $YA=C \Rightarrow Y(AA^{-1})=CA^{-1} \Rightarrow Y=CA^{-1}$

$$Y=\begin{bmatrix}1&2\\3&4\end{bmatrix}\begin{bmatrix}1&-1\\-1&2\end{bmatrix}=\begin{bmatrix}1-2&-1+4\\3-4&-3+8\end{bmatrix}=\begin{bmatrix}-1&3\\-1&5\end{bmatrix}$$

Es. 3

Sia $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

(a) $AX = B$ con $B = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$

(b) $YA = C$ con $C = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$

(a) Da $AX = B$, osservando le dimensioni delle matrici:

$(2,2)(m,n) = (2,2) \Rightarrow m = 2,\ n = 2$ e $X \in M_2(\mathbb{R})$

Studiamo l'esistenza dell'inversa

(1 modo MEG) $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \Rightarrow A$ singolare $\Rightarrow \nexists A^{-1}$

(2 modo Binet-Cauchy)

$\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 4 - 4 = 0 \Rightarrow A$ singolare $\Rightarrow \nexists A^{-1}$

Impostiamo i sistemi risolutivi:

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 6 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cc|cc} x_1 & x_2 & & \\ 1 & 2 & 1 & 3 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

(A = parte sinistra, B = parte destra)

- le # colonne di A indica le # variabili
- le # colonne di B indica le # di sistemi

$\Rightarrow \begin{cases} x_1 + 2x_2 = 1 \\ - \end{cases} \qquad \begin{cases} x_1' + 2x_2' = 3 \\ - \end{cases}$

$\begin{cases} x_1 = 1 - 2x_2 \\ x_2 = t \end{cases} \qquad \begin{cases} x_1' = 3 - 2x_2' \\ x_2' = s \end{cases}$

$\begin{cases} x_1 = 1 - 2t \\ x_2 = t \end{cases} \qquad \begin{cases} x_1' = 3 - 2s \\ x_2' = s \end{cases} \Rightarrow X = \begin{bmatrix} 1-2t & 3-2s \\ t & s \end{bmatrix}$ $t, s \in \mathbb{R}$

test di prova:

$$AX = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1-2t & 3-2s \\ t & s \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} = B$$

(b) Da $YA=C$ dobbiamo passare alla forma $HX=K$.
Passando alle trasposte con $A=\begin{bmatrix}1&2\\2&4\end{bmatrix}$ e $C=\begin{bmatrix}1&3\\2&6\end{bmatrix}$

$$YA=C \Rightarrow (YA)^T=C^T \Rightarrow A^TY^T=C^T \Rightarrow$$

$$\begin{bmatrix}1&2\\2&4\end{bmatrix}Y^T=\begin{bmatrix}1&3\\2&6\end{bmatrix}$$

(A è simmetrica)

Siccome A è singolare $\Rightarrow$ A^T è singolare.
Si calcola Y direttamente.

$$\left[\begin{array}{cc|cc}1&2&1&3\\2&4&2&6\end{array}\right] \xrightarrow{R_2-2R_1} \left[\begin{array}{cc|cc}1&2&1&3\\0&0&0&0\end{array}\right] \Rightarrow$$

$$\begin{cases}x_1+2x_2=1\\ -\end{cases} \qquad \begin{cases}x'_1+2x'_2=3\\ -\end{cases}$$

$$\begin{cases}x_2=t\\ x_1=1-2t\end{cases} \qquad \begin{cases}x'_2=s\\ x'_1=3-2s\end{cases}$$

$$\Rightarrow Y^T=\begin{bmatrix}1-2t & 3-2s\\ t & s\end{bmatrix} \Rightarrow Y=\begin{bmatrix}1-2t & t\\ 3-2s & s\end{bmatrix}$$

test di prova:

$$YA=\begin{bmatrix}1-2t & t\\ 3-2s & s\end{bmatrix}\begin{bmatrix}1&2\\2&4\end{bmatrix}=\begin{bmatrix}1-2t+2t & 2-4t+4t\\ 3-2s+s & 6-4s+4s\end{bmatrix}=\begin{bmatrix}1&2\\3&6\end{bmatrix}=C$$

Es 4

Sia $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ [caso (2,3), rk = 2]

Calcolare

(a) $AX - B$ $\quad B = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$

(b) $YA = C$ $\quad C = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 2 \end{bmatrix}$

(a) Esaminiamo le dimensioni delle matrici in $AX = B$:

$(2,3)(3,4) = (2,2) \Rightarrow 4 = 2 \Rightarrow X \in M_{3,2}$

Impostiamo i sistemi:

$$\left[\begin{array}{ccc|cc} 1 & 2 & 1 & 3 & 5 \\ 0 & 1 & 1 & 1 & 2 \end{array}\right]$$

(x_1 x_2 x_3)

$$\begin{cases} x_1 + 2x_2 + x_3 = 3 \\ x_2 + x_3 = 1 \\ x_3 = s \end{cases} \qquad \begin{cases} x'_1 + 2x'_2 + x'_3 = 5 \\ x'_2 + x'_3 = 2 \\ x'_3 = t \end{cases}$$

$$\begin{cases} x_1 = 2 - 2 + 2s - s \\ x_2 = 1 - s \\ x_3 = s \end{cases} \qquad \begin{cases} x'_1 = -4 + 2t - t + 5 \\ x'_2 = 2 - t \\ x'_3 = t \end{cases}$$

$$\begin{cases} x_1 = 1 + s \\ x_2 = 1 - s \\ x_3 = s \end{cases} \qquad \begin{cases} x'_1 = 1 + t \\ x'_2 = 2 - t \\ x'_3 = t \end{cases}$$

$$X = \begin{bmatrix} 1+s & 1+t \\ 1-s & 2-t \\ s & t \end{bmatrix} \qquad s, t \in \mathbb{R}$$

Test:

$$AX = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1+s & 1+t \\ 1-s & 2-t \\ s & t \end{bmatrix} =$$

$$= \begin{bmatrix} 1+s+2-2s+s & 1+t+4-2t+t \\ 1-s+s & 2-t+t \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix} = B$$

(b) $YA = C$ con $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 2 \end{bmatrix}$.

Esaminiamo le dimensioni delle matrici:

$(n,2)(2,3) = (2,3) \Rightarrow n = 2 \Rightarrow Y \in M_{2,2}(\mathbb{R})$

Passiamo alle Trasposte:

$YA = C \Rightarrow (YA)^T = C^T \Rightarrow A^T Y^T = C^T$.

Determiniamo i sistemi associati:

$$\left[\begin{array}{cc|cc} 1 & 0 & 1 & 1 \\ 2 & 1 & 2 & 3 \\ 1 & 1 & 1 & 2 \end{array}\right] \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \left[\begin{array}{cc|cc} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{array}\right] \xrightarrow{R_3 - R_2} \left[\begin{array}{cc|cc} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$x_1 \quad x_2$

$$\begin{cases} x_1 = 1 \\ x_2 = 0 \end{cases} \quad \begin{cases} x_1' = 1 \\ x_2' = 1 \end{cases} \Rightarrow Y^T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \Rightarrow Y = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

test:

$$YA = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & 2 \end{bmatrix} = C$$

Sia $A=\begin{bmatrix}1&2&3\\2&4&6\end{bmatrix}$ [caso (2,3), rk=1] ES. 5

Calcolare:

(a) $AX=B$ $\quad B=$ [illegible] $\begin{bmatrix}1&2\\2&4\end{bmatrix}$

(b) $YA=C$ $\quad C=\begin{bmatrix}3&6&9\\2&4&6\end{bmatrix}$

(a) Esaminiamo le dimensioni delle matrici:

$AX=B \Rightarrow (2,3)(3,n)=(2,2) \Rightarrow n=2 \Rightarrow X\in M_{3,2}(\mathbb{R})$

Impostiamo i sistemi:

$$\left[\begin{array}{ccc|cc}1&2&3&1&2\\2&4&6&2&4\end{array}\right] \xrightarrow{R_2-2R_1} \left[\begin{array}{ccc|cc}1&2&3&1&2\\0&0&0&0&0\\x_1&x_2&x_3&&\end{array}\right]$$

$$\begin{cases}x_1+2x_2+3x_3=1\\x_2=\pi\\x_3=\sigma\end{cases} \qquad \begin{cases}x_1'+2x_2'+3x_3'=2\\x_2'=\pi'\\x_3'=\sigma'\end{cases}$$

$$\begin{cases}x_1=1-2x_2-3x_3=1-2\pi-3\sigma\\x_2=\pi\\x_3=\sigma\end{cases} \qquad \begin{cases}x_1'=2-2x_2'-3x_3'=2-2\pi'-3\sigma'\\x_2'=\pi'\\x_3'=\sigma'\end{cases}$$

$$X=\begin{bmatrix}1-2\pi-3\sigma & 2-2\pi'-3\sigma'\\ \pi & \pi'\\ \sigma & \sigma'\end{bmatrix} \qquad \pi,\sigma,\pi',\sigma'\in\mathbb{R}$$

test:

$$AX=\begin{bmatrix}1&2&3\\2&4&6\end{bmatrix}\begin{bmatrix}1-2\pi-3\sigma & 2-2\pi'-3\sigma'\\ \pi & \pi'\\ \sigma & \sigma'\end{bmatrix}=$$

$$=\begin{bmatrix}1-2\pi-3\sigma+2\pi+3\sigma & 2-2\pi'-3\sigma'+2\pi'+3\sigma'\\ 2-4\pi-6\sigma+4\pi+6\sigma & 4-4\pi'-6\sigma'+4\pi'+6\sigma'\end{bmatrix}=$$

$$=\begin{bmatrix}1&2\\2&4\end{bmatrix}=B$$

⑥ Esaminiamo le dimensioni in $YA = C$

ove $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$ e $C = \begin{bmatrix} 3 & 6 & 9 \\ 2 & 4 & 6 \end{bmatrix}$

$(h,2)(2,3) = (2,3) \Rightarrow n = 2 \Rightarrow Y \in M_2(\mathbb{R})$.

Passiamo alle Trasposte:

$YA = C \Rightarrow (YA)^T = C^T \Rightarrow A^T Y^T = C^T$

Determiniamo i sistemi associati:

$$\left[\begin{array}{cc|cc} 1 & 2 & 3 & 2 \\ 2 & 4 & 6 & 4 \\ 3 & 6 & 9 & 6 \end{array}\right] \xrightarrow[R_3 - 3R_1]{R_2 - 2R_1} \left[\begin{array}{cc|cc} 1 & 2 & 3 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow$$

$$\begin{cases} x_1 + 2x_2 = 3 \\ x_2 = s \end{cases} \qquad \begin{cases} x_1' + 2x_2' = 2 \\ x_2' = t \end{cases}$$

$$\begin{cases} x_1 = 3 - 2s \\ x_2 = s \end{cases} \qquad \begin{cases} x_1' = 2 - 2t \\ x_2' = t \end{cases} \Rightarrow Y^T = \begin{bmatrix} 3-2s & 2-2t \\ s & t \end{bmatrix} \Rightarrow$$

$$Y = \begin{bmatrix} 3-2s & s \\ 2-2t & t \end{bmatrix} \qquad s, t \in \mathbb{R}$$

<u>test</u>:

$$YA = \begin{bmatrix} 3-2s & s \\ 2-2t & t \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} =$$

$$= \begin{bmatrix} 3-2s+2s & 6-4s+4s & 9-6s+6s \\ 2-2t+2t & 4-4t+4t & 6-6t+6t \end{bmatrix} = \begin{bmatrix} 3 & 6 & 9 \\ 2 & 4 & 6 \end{bmatrix} = C$$

Sia $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$ [corso (3, 2), n.to 2] 6

Calcolare:

(a) $AX = B \quad B = \begin{bmatrix} 2 & 1 \\ 5 & 1 \\ 3 & 0 \end{bmatrix}$

(b) $YA = C \quad C = \begin{bmatrix} 4 & 3 \\ 2 & 6 \end{bmatrix}$

(a) Esaminiamo le dimensioni in $AX = B$

$(3,2)(2,n) = (3,2) \Rightarrow n = 2 \Rightarrow X \in M_2(\mathbb{R})$

Impostiamo i sistemi:

$$\left[\begin{array}{cc|cc} 1 & 0 & 2 & 1 \\ 1 & 1 & 5 & 1 \\ 0 & 1 & 3 & 0 \end{array}\right] \xrightarrow{R_2 - R_1} \left[\begin{array}{cc|cc} 1 & 0 & 2 & 1 \\ 0 & 1 & 3 & 0 \\ 0 & 1 & 3 & 0 \end{array}\right] \xrightarrow{R_3 - R_2} \left[\begin{array}{cc|cc} 1 & 0 & 2 & 1 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\begin{cases} x_1 = 2 \\ x_2 = 3 \end{cases} \quad \begin{cases} x'_1 = 1 \\ x'_2 = 0 \end{cases} \Rightarrow X = \begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix}$$

Verifica:

$$AX = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 2+3 & 1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 5 & 1 \\ 3 & 0 \end{bmatrix} = B$$

(b) Esaminiamo le dimensioni in $YA = C$

ove $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$, $C = \begin{bmatrix} 4 & 3 \\ 2 & 6 \end{bmatrix}$

$(n,3)(3,2) = (2,2) \Rightarrow n = 2 \Rightarrow Y \in M_{2,3}(\mathbb{R})$

~~Impostazione sistemi risolvendo una matrice~~

Passiamo alla trasposta

$YA = C \Rightarrow (YA)^T = C^T \Rightarrow A^T Y^T = C^T$

Impostiamo i sistemi:

$$\left[\begin{array}{ccc|cc} 1 & 1 & 0 & 4 & 2 \\ 0 & 1 & 1 & 3 & 6 \end{array}\right] \Rightarrow$$

(x_1 x_2 x_3)

$$\begin{cases} x_1 + x_2 = 4 \\ x_2 + x_3 = 3 \\ x_3 = s \end{cases} \qquad \begin{cases} x'_1 + x'_2 = 2 \\ x'_2 + x'_3 = 6 \\ x'_3 = t \end{cases}$$

$$\begin{cases} x_1 = 4 - 3 + s = 1 + s \\ x_2 = 3 - s \\ x_3 = s \end{cases} \qquad \begin{cases} x'_1 = 2 - 6 + t = -4 + t \\ x'_2 = 6 - t \\ x_3 = t \end{cases} \Rightarrow Y^T = \begin{bmatrix} 1+s & t-4 \\ 3-s & 6-t \\ s & t \end{bmatrix}$$

$$\Rightarrow Y = \begin{bmatrix} 1+s & 3-s & s \\ t-4 & 6-t & t \end{bmatrix}$$

Verifica

$$YA = \begin{bmatrix} 1+s & 3-s & s \\ t-4 & 6-t & t \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} =$$

$$= \begin{bmatrix} 1+s+3-s & 3-s+s \\ t-4+6-t & 6-t+t \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 2 & 6 \end{bmatrix} = C$$

Sia $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 1 & 2 \end{bmatrix}$ ... [caso (3,2), rk 1] ... Es. 7

Calcolare:

(a) $AX = B \qquad B = \begin{bmatrix} 3 & 5 \\ 6 & 10 \\ 3 & 5 \end{bmatrix}$

(b) $YA = C \qquad C = \begin{bmatrix} 4 & 8 \\ 2 & 4 \end{bmatrix}$

(a) $AX = B$. Esaminiamo le dimensioni:

$(3,2)(2,n) = (3,2) \Rightarrow n = 2 \Rightarrow X \in M_2(\mathbb{R})$.

Determiniamo i sistemi omogenei:

$$\left[\begin{array}{cc|cc} 1 & 2 & 3 & 5 \\ 2 & 4 & 6 & 10 \\ 1 & 2 & 3 & 5 \end{array}\right] \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \left[\begin{array}{cc|cc} 1 & 2 & 3 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow \begin{array}{l} rk = 1 \\ \#\text{incognite} = 2 \\ \#\text{sistemi} = 2 \end{array}$$

$$\begin{cases} x_1 + 2x_2 = 3 \\ x_2 = x \end{cases} \qquad \begin{cases} x'_1 + 2x'_2 = 5 \\ x'_2 = t \end{cases}$$

$$\begin{cases} x_1 = 3 - 2x \\ x_2 = x \end{cases} \qquad \begin{cases} x'_1 = 5 - 2t \\ x'_2 = t \end{cases} \Rightarrow X = \begin{bmatrix} 3 - 2x & 5 - 2t \\ x & t \end{bmatrix} \quad x, t \in \mathbb{R}$$

Verifica:

$$AX = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 3 - 2x & 5 - 2t \\ x & t \end{bmatrix} =$$

$$= \begin{bmatrix} 3 - 2x + 2x & 5 - 2t + 2t \\ 6 - 4x + 4x & 10 - 4t + 4t \\ 3 - 2x + 2x & 5 - 2t + 2t \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 6 & 10 \\ 3 & 5 \end{bmatrix} = B$$

(5) Esaminiamo le dimensioni in $YA=C$

con $C=\begin{bmatrix}4 & 8\\ 2 & 4\end{bmatrix} \Rightarrow (4,3)(3,2)=(2,2) \Rightarrow 4=2 \Rightarrow Y\in M_{2,3}$

Passiamo alla trasposta:

$YA=C \Rightarrow (YA)^T=C^T \Rightarrow A^TY^T=C^T$

Determiniamo i sistemi ausiliari:

$$\left[\begin{array}{ccc|cc} 1 & 2 & 1 & 4 & 2\\ 2 & 4 & 2 & 8 & 4\end{array}\right] \xrightarrow{R_2-2R_1} \left[\begin{array}{ccc|cc} 1 & 2 & 1 & 4 & 2\\ 0 & 0 & 0 & 0 & 0\end{array}\right]$$

$$\begin{cases} x_1+2x_2+x_3=4\\ x_2=x\\ x_3=t\end{cases} \qquad \begin{cases} x'_1+2x'_2+x'_3=2\\ x'_2=x'\\ x'_3=t'\end{cases}$$

$$\begin{cases} x_1=4-2x-t\\ x_2=x\\ x_3=t\end{cases} \qquad \begin{cases} x'_1=2-2x'-t'\\ x'_2=x'\\ x'_3=t'\end{cases} \Rightarrow$$

$$Y^T=\begin{bmatrix} 4-2x-t & 2-2x'-t'\\ x & x'\\ t & t'\end{bmatrix}$$

$$\Rightarrow Y=\begin{bmatrix} 4-2x-t & x & t\\ 2-2x'-t' & x' & t'\end{bmatrix} \qquad x,t,x',t'\in\mathbb{R}.$$

Verifica:

$$YA=\begin{bmatrix} 4-2x-t & x & t\\ 2-2x'-t' & x' & t'\end{bmatrix}\begin{bmatrix} 1 & 2\\ 2 & 4\\ 1 & 2\end{bmatrix}=$$

$$=\begin{bmatrix} 4-2x-t+2x+t & 8-4x-2t+4x+2t\\ 2-2x'-t'+2x'+t' & 4-4x'-2t'+4x'+2t'\end{bmatrix}=\begin{bmatrix} 4 & 8\\ 2 & 4\end{bmatrix}=C$$